BEFORE YOU READ

Where permutation matrices live

Permutation matrices sit inside a chain of matrix classes: a few are elementary row swaps, all of them are orthogonal, and every orthogonal matrix is invertible.

$$E \subset P \subset Q \subset A$$

A **Invertible**

A exists

Q **Orthogonal**

$Q^{-1} = Q^{T}$

P **Permutation** `n! of them`

adds any reordering · one 1 in each row and column

E **Row-swap elementary**

swaps a single pair of rows

TAKEAWAY

Inverting is free

Because permutation matrices are orthogonal, $P^{-1} = P^{T}$: undoing a permutation means transposing it.